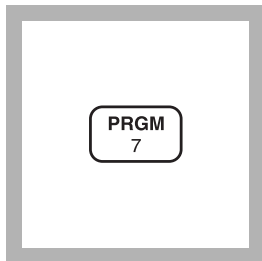


SULFIDE (0 to 0.70 mg/L S²⁻)

For water, wastewater and seawater

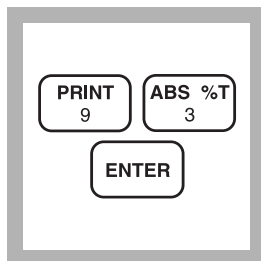
Methylene Blue Method* USEPA accepted for reporting wastewater analysis**

1. Enter the stored program number for sulfide (S).

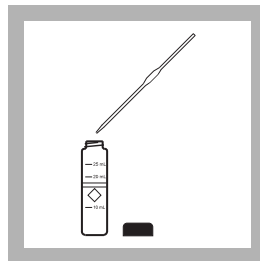
Press: **PRGM**

The display will show:

PRGM ?



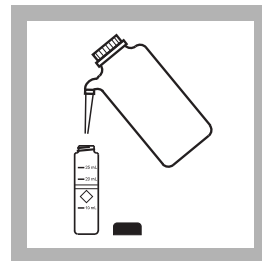
2. Press: **93 ENTER**
The display will show **mg/L, S** and the **ZERO** icon.



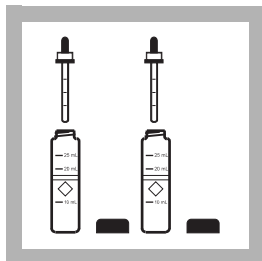
3. Pipet 25 mL of sample into a clean sample cell. This will be the prepared sample.

Note: Samples must be analyzed immediately and cannot be preserved for later analysis. Use a pipet to avoid agitation.

Note: For field testing, a 25-mL graduated cylinder may be used.

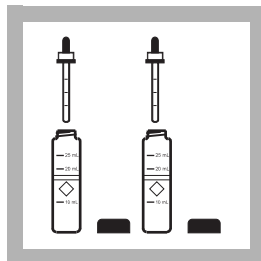


4. Fill a second sample cell with 25 mL of deionized water (the blank).



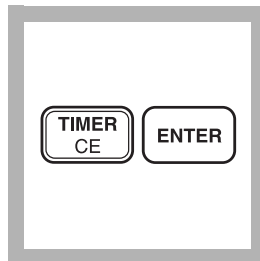
5. Add 1.0 mL of Sulfide 1 Reagent to each cell. Swirl to mix.

Note: Use the calibrated 1-mL dropper.

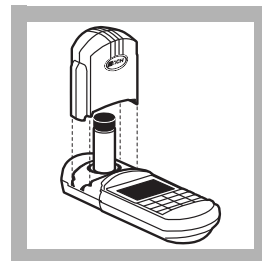


6. Add 1.0 mL of Sulfide 2 Reagent to each cell. Immediately swirl to mix.

Note: A pink color will develop, then the solution will turn blue if sulfide is present.



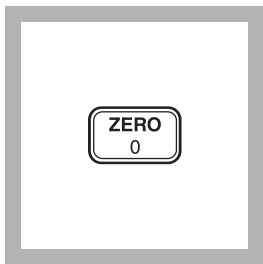
7. Press:
TIMER ENTER
A 5-minute reaction period will begin.



8. Place the blank into the cell holder. Tightly cover the sample cell with the instrument cap.

* Adapted from *Standard Methods for the Examination of Water and Wastewater*.

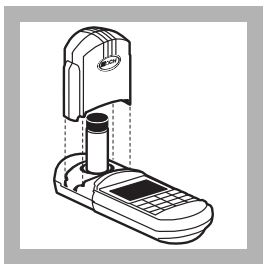
** Procedure is equivalent to USEPA method 376.2 or Standard Method 4500-S²⁻ D for wastewater.



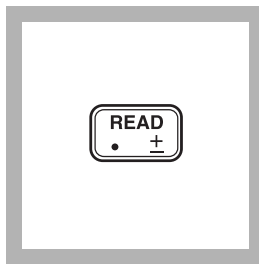
9. Press: ZERO

The cursor will move to the right, then the display will show:

0.00 mg/L S



10. After the timer beeps, place the prepared sample into the cell holder. Tightly cover the sample cell with the instrument cap.



11. Press: READ

The cursor will move to the right, then the result in mg/L sulfide will be displayed.

***Note:** Some sulfide loss may occur if dilution is necessary.*

***Note:** Standard Adjust may be performed using a prepared standard (see Standard Adjust in Section 1).*

Sampling and Storage

Collect samples in clean plastic or glass bottles. Fill completely and cap tightly. Avoid excessive agitation or prolonged exposure to air. Analyze samples immediately.

Accuracy Check

Sulfide standards are unstable and must be user prepared. See Standard Methods, 4500S⁻ for preparation and standardization directions.

Method Performance

Precision

In a single laboratory, using standard solutions of 0.73 mg/L sulfide and two representative lots of reagent with the instrument, a single operator obtained a standard deviation of ± 0.02 mg/L sulfide.

Estimated Detection Limit (EDL)

The EDL for program 93 is 0.01 mg/L S²⁻. For more information on derivation and use of Hach's estimated detection limit, see Section 1.

Interferences

Interfering Substance	Interference Levels and Treatments
Strong reducing substances (sulfite, thiosulfate and hydrosulfite)	Interfere by reducing the blue color or preventing its development.
Sulfide, high levels	High concentrations of sulfide may inhibit full color development and require sample dilution. Some sulfide loss may occur when the sample is diluted.
Turbidity	For turbid samples, prepare a sulfide-free blank as follows. Use it in place of the deionized water blank in the procedure. 1. Measure 25 mL of sample into a 50-mL erlenmeyer flask. 2. Add Bromine Water dropwise with constant swirling until a permanent yellow color just appears. 3. Add Phenol Solution dropwise until the yellow color just disappears. Use this solution in Step 4 in place of deionized water.

Soluble Sulfides

Determine soluble sulfides by centrifuging the sample in completely filled, capped tubes and analyzing the supernatant. Insoluble sulfides are then estimated by subtracting the soluble sulfide concentration from the total sulfide result.

Summary of Method

Hydrogen sulfide and acid-soluble metal sulfides react with N, N-dimethyl-p-phenylenediamine oxalate to form methylene blue. The intensity of the blue color is proportional to the sulfide concentration. High sulfide levels in oil field waters may be determined after dilution.

Pollution Prevention and Waste Management

Sulfide 2 Reagent contains potassium dichromate. The final solution will contain hexavalent chromium (D007) at a concentration regulated as a hazardous waste by Federal RCRA. See *Section 3* for more information on proper disposal of these materials.

SULFIDE, continued

REQUIRED REAGENTS

	Cat. No.
Sulfide Reagent Set (100 tests).....	22445-00
Includes: (2) 1816-42, (2) 1817-42	

Description	Quantity Required		Units	Cat. No.
	Per Test			
Sulfide 1 Reagent.....	2 mL.....	100 mL	MDB.....	1816-32
Sulfide 2 Reagent.....	2 mL.....	100 mL	MDB.....	1817-32
Water, deionized.....	25 mL.....	4 L		272-56

REQUIRED APPARATUS

Cylinder, graduated, 25 mL	1	each.....	508-40
Pipet, volumetric, Class A, 25.00 mL.....	1	each.....	14515-40
Pipet Filler, safety bulb	1	each.....	14651-00
Sample Cell, 10-20-25 mL, w/ cap	2	6/pkg.....	24019-06

OPTIONAL REAGENTS

Description	Units	Cat. No.
Bromine Water, 30 g/L.....	29 mL.....	2211-20
Phenol Solution, 30 g/L	29 mL.....	2112-20
Sodium Sulfide, hydrate	114 g.....	785-14

OPTIONAL APPARATUS

Bottle, Wash, 250 mL	each.....	620-31
Dropper, for 1 oz. bottle.....	each.....	2258-00
Flask, erlenmeyer, 50 mL	each.....	505-41
<i>Standard Methods for the Examination of Water and Wastewater</i>	each.....	22708-00

For Technical Assistance, Price and Ordering

In the U.S.A.—Call 800-227-4224

Outside the U.S.A.—Contact the Hach office or distributor serving you.